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SUPERSEDING
MIL-G-7601A(ASG)
4 January 1954

MILITARY SPECIFICATION

GAGE, EMERGENCY OXYGEN, TYPE L-2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

*1.1 Scope. This specification covers one type of high-pressure, emergency oxygen gage, range 0 to 2,500 pounds, designated Type L-2.

*2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

*	BB-N-411	Nitrogen, Technical
*	BB-O-925	Oxygen, Technical, Gas and Liquid

Military

	MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT General Requirements for
	MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
	MIL-T-10727	Tin Plating; Electrodeposited or Hot-Dipped, for Ferrous and Nonferrous Metals
*	DOD-D-1000	Drawing, Engineering and Associated Lists
*	MIL-C-52211	Components and Assemblies for Industrial Gas Production, Storage and Transport Equipment, Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, SA-ALC/MMEDO, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
* MIL-STD-889	Dissimilar Metals

Air Force-Navy Aeronautical Design Standards

AND10052	Boss Standard Dimensions for External Pipe Thread

*(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Consolidated Classification Committee

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606).

*2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

*3.1 Preproduction. The gage furnished under this specification shall be a product which has been tested and has passed the preproduction tests specified herein.

3.2 Materials.

3.2.1 Nonferrous materials. Unless otherwise specified, nonferrous materials shall be used for all parts of the gage except where ferrous materials are essential. Ferrous materials, when used, shall not come in direct contact with oxygen.

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*3.2.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.2.2 Nonmetallic materials. Any nonmetallic material that is easily deteriorated or otherwise easily affected adversely by continued service use with oxygen shall not be used.

3.2.3 Protective treatment. When materials are used in the construction of the gage that are subject to corrosion in oxygen, salt air, or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coatings and finishes that will crack, chip, or scale with normal service life, age, or extremes of atmospheric conditions shall not be used.

*3.2.4 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143.

3.3 Design. The gage shall be designed to conform to Figure 1.

3.4 Construction. The gages shall be so constructed that no parts will work loose in service. The mechanism shall be built to withstand without failure, the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. Pivots, bearings, and gears, shall neither bind nor shake, and shall be as near frictionless as practicable. The gages shall be so constructed that minor repairs such as replacement of cover glasses and pointers can be easily made by the personnel of the operating units without the use of special tools.

3.4.1 Hair springs. If hair springs are used in the construction of the gage, they shall be made of beryllium copper or phosphor bronze.

3.4.2 Bourdon tube. The bourdon tube shall be constructed of beryllium copper.

3.4.3 Cover glass. The cover glass shall be transparent plastic material, clear, flat, and free from flaws which will affect the readability of the gage. The plastic material shall not craze and shall remain transparent when subjected to the temperatures specified herein.

3.4.4 Case.

3.4.4.1 Base. The base of the case shall be fabricated from brass or other suitable metal.

3.4.4.2 Cover. The cover of the case shall be fabricated from steel or other material of suitable strength to withstand the normal service conditions. Materials such as soft aluminum which are subject to damage under service conditions shall not be used.

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*3.4.5 Protective cap. The external pipe thread fitting conforming to Standard AND10052 shall be fitted with a suitable cap to protect the threads and to prevent dust and foreign matter from entering the gage during shipping and storage.

3.4.6 Pointer. The pointer shall be of sufficient length that the top end will extend well into the marked surface of the dial. The pointer shall be dull black in color.

3.4.6.1 Pointer adjustments. The pointer shall be 0.020 ± 0.005 inch wide at the point where it crosses the lower scale arc. The pointer shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration. A mechanical high limit stop, if used, shall be applied to the mechanism and not to the pointer. The pointer shall be firmly attached to the mechanism, yet shall be readily adjustable.

3.4.7 Dial. The dial shall be securely fastened upon the frame mechanism, in order that it will not become loose or turn when the gage is vibrated.

3.4.8 Dial to cover glass distance. The distance between the dial and the inside of the cover glass shall be the minimum practicable, but shall be sufficient to insure freedom of movement of the pointer under all conditions of gage operation.

3.4.9 Mechanism adjustment. The mechanism shall be provided with means to adjust or correct the indications, if this becomes necessary, during the use or service life of the gage. This means of adjustment shall be reasonably simple to manipulate, and of such a nature that adjustment can be accomplished without the use of special tools.

3.4.10 External construction. The indicating and pressure elements shall comprise a unit distinct from the case. This unit shall be sufficiently rugged in order that small distortions of the case will not affect the accuracy of the indications.

*3.5 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000.

3.6 Weight. The weight of the gage shall not exceed 0.16 pound.

3.7 Threads.

3.7.1 Screw. Machine screw threads shall be in accordance with Specification MIL-S-7742.

*3.7.2 Pipe. Pipe threads shall be in accordance with Specification MIL-P-7105. Threads shall be tinned in accordance with Specification MIL-T-10727.

3.8 Filter. The inlet shall be provided with a sintered bronze porous plug to prevent foreign materials entering the gage.

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3.9 Performance.

3.9.1 Leakage.

3.9.1.1 Bourdon tube. The Bourdon tube and base, upon completion of the assembly, shall be capable of withstanding a gas pressure of 2,000 psi while immersed in a water bath without leakage.

3.9.1.2 Case. The case shall be mechanically tight in all joints.

3.9.2 Scale error. The master gage of known accuracy shall indicate at least 1,800-psi pressure, but not more than 1,900-psi pressure, when the gage being tested indicates 1,800-psi pressure.

3.9.3 Friction. The pointer shall not move more than the width of the black indication marker when a pressure of 1,800 psi is applied and the gage tapped lightly. The pointer shall move smoothly throughout the complete range without vibration.

3.9.4 Drift. The pointer of the gage shall not move out of the width of the black indication marker at 1,800 pounds after being subjected to 1,800-psi pressure for 1 hour. The gage shall be lightly tapped during this test.

3.9.5 Seasoning. The gage shall be capable of satisfactory operation after being subjected to 300 applications of 1,800-psi pressure.

3.9.6 Temperature.

3.9.6.1 Low temperature. The gage shall function properly at a temperature of -65°F.

3.9.6.2 High temperature. The gage shall function properly at a temperature of +160°F.

3.9.7 Overpressure. The gage shall not leak or burst while being subjected to a hydrostatic pressure of 3,500 psi for 5 minutes.

3.9.8 Vibration. The gage shall function satisfactorily after being vibrated at any desired frequency between 500 and 2,500 cpm and a double amplitude of not less than 0.018 nor more than 0.020 inch for a period of 3 hours.

3.10 Markings.

3.10.1 Dial markings. The dial shall have a white background with dull black markings, unless otherwise specified. It shall be marked with an arc 1/32 inch wide showing the pressure range of the gage. A black indication marker, consisting of a segment of a width corresponding to 100-psi segment of the scale arc plus the width of the pointer at the scale arc, shall be placed at the point corresponding to the 1,800-psi pressure indication on the arc. This marker shall be set normal to the upper scale arc surface. The true 1,800-psi pressure on the gage shall be indicated on the gage by the left-hand edge of the pointer lining up with the left-hand edge of the 1,800-pound indicator marking. The 1,800-pound indication shall be plainly marked on the dial.

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3.10.1.1 The remainder of the indicator scale on the up-pressure side shall be marked in black and continue to the 2,500-pound pressure limit. This pressure reading shall be plainly marked at its up-pressure limit end.

*3.10.1.2 The indicator scale, starting at the left edge of the 1,800-pound mark shall be continued to the zero pressure limit in a lusterless red color, in accordance with shade No. 31136 of FED-STD-595. The words "refill" shall be in red, and "full" in black and shall be located in the arc as shown on Figure 1. The word "oxygen" shall be marked on the dial in black. "Full" and "Refill" may be located on the opposite side of the arc instead of as shown on Figure 1.

3.10.1.3 The following marking in red letters shall be placed on the dial, as shown on Figure 1: "Use No Oil."

3.10.1.4 All marking shall be durable to withstand usage encountered in service.

3.11 Identification of product. Each gage shall be marked in accordance with Standard MIL-STD-130, with the following information, using letters approximately 1/16 inch high:

GAGE, EMERGENCY OXYGEN
Type L-2
Specification MIL-G-7601B
US

*3.11.1 Use of AN, MIL or MS designations. AN, MIL or MS designations shall not be applied to a product, except for preproduction test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for preproduction tests.

3.12 Workmanship. Workmanship shall be in accordance with high-grade aircraft instrument manufacturing practice. The gage shall be thoroughly cleaned of all oil and foreign matter at the proper stage of assembly.

4. QUALITY ASSURANCE PROVISIONS.

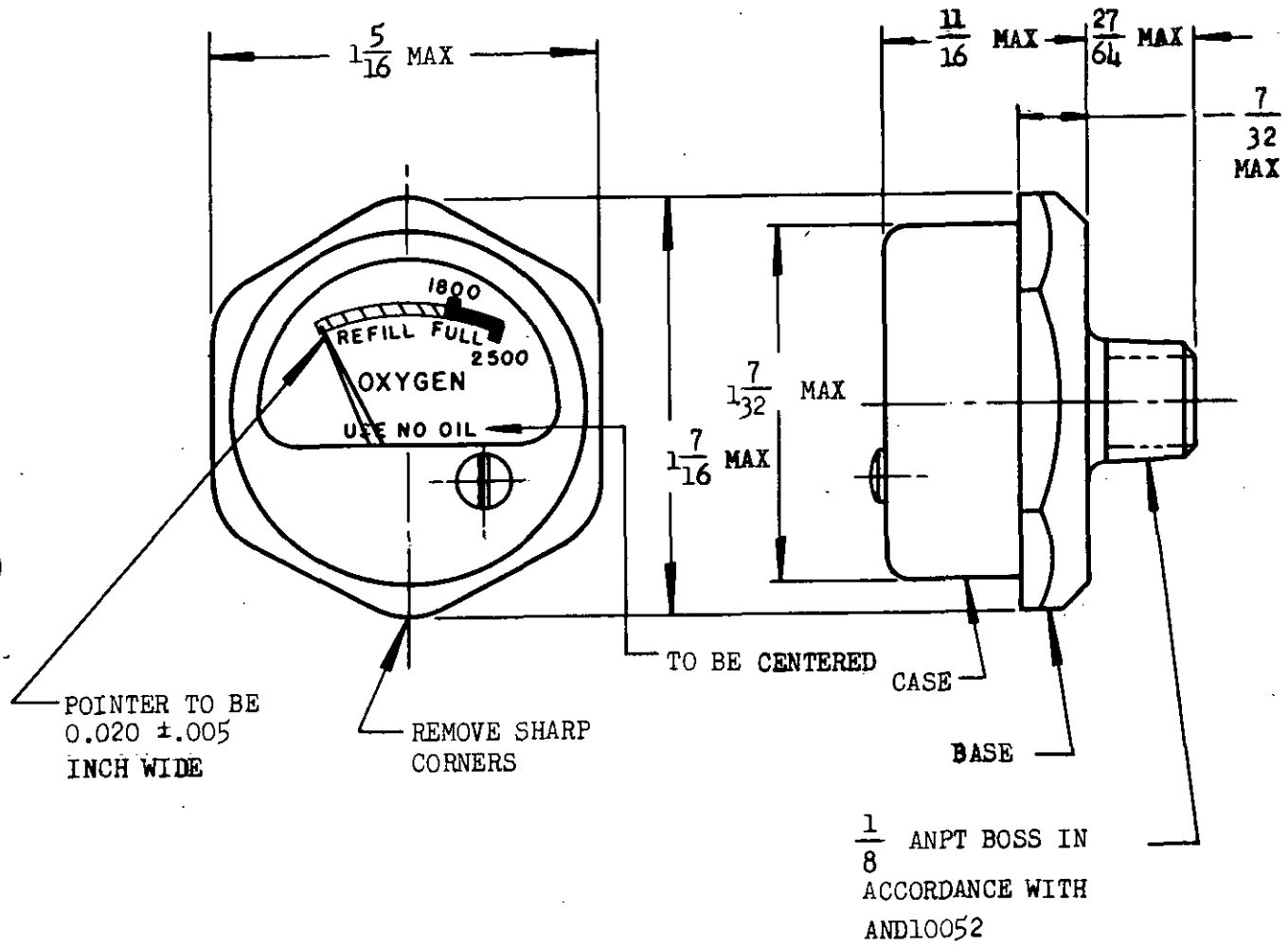
*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1.1 Classification of tests. The inspection and testing of gages shall be classified as follows:

(a) Preproduction tests: Preproduction tests are tests performed on samples in accordance with the tests specified herein (see 4.3).

(b) Quality conformance tests: Conformance tests are those tests performed on individual lots which have been submitted for conformance.

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DIAL MARKINGS TO BE 1/16 INCH HIGH MIN
1/8 ANPT SHALL BE TINNED IN ACCORDANCE WITH SPECIFICATION MIL-T-10727
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES FRACTIONS ±1/64

FIGURE 1. Gage

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4.2 Test conditions.

4.2.1 Atmospheric pressure and temperature. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and a temperature of approximately 68°F. When tests are made with ambient pressure or temperature different from the above values, proper allowance shall be made for the difference from the specified conditions.

*4.2.2 Gas. The high-pressure gas used in testing the gage shall be either oxygen conforming to Specification BB-O-925, water-pumped nitrogen confroming to Specification BB-N-411, or water-pumped air equivalent in dryness to Specification BB-N-411 nitrogen.

4.2.3 Tapping. Unless otherwise specified, the gage shall be tested in normal vertical oeprating position and shall be lightly tapped before a test reading is taken.

4.2.4 Vibration. Vibration shall be at any desired frequency between 500 and 2,500 cpm and at a double amplitude of not less than 0.018 nor more than 0.020 inch.

4.2.5 Tolerance. To provide for changes in the gage mechanism due to expansion and contraction of the parts when testing for scale errors under conditions of high and low temperatures specified herein, the following deviation from scale error test will be allowed. During the rechecking for the high and low temperature scale error tests, a deviation will be allowed whereby the master gage shall indicate at least 1,800 psi but not more than 1,950 psi.

*4.3 Preproduction tests.

4.3.1 Preproduction test. Preproduction test samples shall consist of four gages to be tested in accordance with test specified herein.

*4.3.2 Tests. The preproduction tests of gages shall consist of all the tests specified under quality conformance tests.

4.4 Quality conformance tests. Quality conformance tests shall consist of Individual tests and Sampling tests.

4.4.1 Individual tests. Each gage shall be subjected to the following tests.

4.4.1.1 Examination of product. Unless otherwise specified by the procuring activity, each gage shall be examined to determine conformance with this specification with respect to materials, workmanship, and markings.

4.4.1.2 Leakage tests.

4.4.1.2.1 Bourdon tube and base. After assembly of the Bourdon tube and base and before attachment of the dial and pointer, each such assembly shall be tested at a gas pressure of 2,000 psi to insure absolute freedom from leaks as evidenced by no bubbles.

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4.4.1.3 Scale error. The gage being tested and a master test gage of known accuracy shall be connected in parallel and subjected to a source of pressure, and pressure applied until the pointer of the sample gage rests upon the 1,800-psi indication. The master gage at this point shall not read less than 1,800 psi, nor more than 1,900 psi. The pressure on the master gage shall be increased to a reading of 2,000 pounds. The pressure shall then be bled down until the pointer, on the gage being tested, again indicates 1,800 psi. At this point, the reading on the master gage shall not exceed 1,900 pounds, nor be less than 1,800 pounds.

4.4.1.4 Friction. The gage shall be tested for friction at 1,800 pounds in the following manner. The pressure shall be increased on the gage until the pointer reaches its correct position for showing 1,800-psi pressure; at this point the gage shall be tapped and the pointer shall not move out of the width of the 1,800-pound graduation on the scale. The pointer shall move smoothly while the pressure is varied uniformly without vibration of the gage. This test may be combined with the test for scale error.

4.4.2 Sampling tests. Samples selected as specified in plans A and B shall be subjected to the applicable tests. A lot shall consist of all gages manufactured under essentially the same conditions at substantially the same time.

4.4.2.1 Sampling plan A. Five gages shall be selected at random from each lot and subjected to the following tests.

4.4.2.1.1 Vibration and low temperature. The gage shall be mounted on a suitably designed vibration stand and subjected to a temperature of -65°F. While at this temperature, the gage shall be subjected to vibration for a period of 3 hours, duplicating as nearly as practicable an actual installation. The vibration shall be as specified herein and at a frequency of from 500 to 2,500 cpm. During this test, a gas pressure of 1,800 psi minimum shall be maintained on the gage.

4.4.2.1.1.1 The gage shall be tested for scale errors as specified in the Individual tests, after the 3-hour period and after vibration has been stopped but the gage is still at the specified temperature. The tolerance specified in paragraph 4.2.5 shall apply during this test.

4.4.2.1.1.2 After the preceding scale error and vibration at low temperature tests, the gage shall be tested for leaks as specified herein.

4.4.2.1.2 High temperature. The gage shall be subjected to a temperature of +180°F for a period of 3 hours, after which it shall again be tested for scale error and leakage as specified herein. The tolerances specified in paragraph 4.2.5 shall apply during this test.

4.4.2.1.3 Drift. The gage shall be subjected to a pressure of 1,800 psi for 1 hour, after which the reading of the gage at 1,800 pounds shall not have increased by more than the width of the 1,800-psi graduation.

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4.4.2.1.4 High pressure. The gage being tested and a master gage of known accuracy shall be connected in parallel and subjected to a source of pressure, and pressure applied until the pointer of the sample gage rests upon the 1,800-psi indication. The master gage at this point shall not read less than 1,800 psi, nor more than 1,900 psi. The pressure on the master gage shall be increased to a reading of 2,500 psi. The pressure shall then be bled down until the pointer, on the gage being tested, again indicates 1,800 psi. At this point, reading on the master gage shall not exceed 1,950 pounds, nor be less than 1,800 pounds.

4.4.2.2 Sampling plan B. One gage shall be selected at random from each lot and subjected to the following tests.

4.4.2.2.1 Overpressure. The gage shall be subjected to a hydrostatic over-pressure test of 3,500 psi for a period of 5 minutes without bursting. The gage shall then be thoroughly dried and again subjected to the Leakage test. Gages subjected to this test shall not be submitted as part of the contract or order.

4.4.2.2.2 Seasoning. The gage shall be subjected to 300 applications of pressure from 0 to 1,800 psi and back to 0 psi at a frequency of approximately 1 cpm. Twenty-four hours after the completion of this test, the gage shall satisfactorily pass the Scale error test.

*4.4.3 Rejection. Any gage failing to meet the requirements of the Individual tests shall be rejected and returned at the contractor's expense. When any sample fails to meet the requirements of the Sampling tests, the lot represented shall be rejected and returned at the contractor's expense.

4.4.3.1 When tests are specified on a quantity of gages that are selected as representative of a certain lot, and one or more of this number fails to meet the requirements, additional gages of the lot represented shall be tested immediately to determine the cause of failure. Individual performance tests shall not be interrupted, unless the defect is of such a nature that it will seriously affect the performance or safe use of the gage.

5. PACKAGING.

5.1 Preservation and Packaging. Preservation and packaging shall be in accordance with MIL-C-52211. Method optional.

5.2 Packing. Packing shall be level A, B, C or industrial in accordance with MIL-C-52211, as specified (see 6.2).

5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows: Gage, Emergency Oxygen, Type L-2, Manufacturer's Part No. * , Specification MIL-G-7601B, FRAGILE, DELICATE INSTRUMENTS, HANDLE WITH CARE, Federal Stock No. * (if no FSN available, leave space therefor).

*Information to be entered by the manufacturer.

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6. NOTES.

6.1 Intended use. The Type L-2 emergency oxygen gage covered by this specification is used with the Type H-2 cylinder assembly and is intended principally to show whether or not the cylinder is fully charged and is also used to read accurately at the 1,800-psi mark.

*6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether domestic or overseas packing is required (see section 5).

* (c) Applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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